AIAA Propulsion and Energy Forum
August 9-11, 2021, VIRTUAL EVENT
AIAA Propulsion and Energy 2021 Forum

10.2514/6.2021-3501

Numerical Simulation of Centrifugal Casting of Paraffin Wax: Influence of Rotational Speed and Fluid Viscosity

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Paraffin wax as an alternative hybrid rocket propellant has received a lot of attention from researchers because of its attractive characteristics; high-performing, non-toxic, reusable, and easy to store. Paraffin wax fuel grains are normally manufactured through a centrifugal casting process. In the centrifugal casting process, fluid properties such as viscosity and casting tube geometrical parameters have a direct impact on the quality of the final cast product. In order to further understand and predict various heat and mass flow phenomena, and even optimise the centrifugal casting process, it is necessary to study fluid flow patterns in the centrifugal casting process. This research paper numerically investigates the influence of rotational speed and fluid viscosity in the centrifugal casting process of paraffin wax. Finite volume code in the Computational Fluid Dynamics (CFD) solver ANSYS Fluent was employed for the simulations. Liquid viscosity (water, SAE 5W-30 motor oil, and paraffin wax dotriacontane $C_{32}H_{66}$), and rotational speeds (1000 RPM, 2220 RPM, and 3000 RPM) were studied. From the transient two-phase isothermal simulations, it was noted that viscosity played a major role in the centrifugal casting process; with high viscosity liquids like SAE 5W-30 motor oil forming a smooth, reliable annulus much faster, and even at relatively lower rotational speeds, unlike low viscosity liquids like water.

I. Nomenclature

d	Diameter $[m]$
g	Gravitational Acceleration $[m/s^2]$
h	Coating Thickness of Rotating Fluid $[m]$
l	Length $[m]$
N	Revolutions Per Minute [-]
r	Radius $[m]$
v	Linear Velocity $[m/s]$
Greek Symbols	
μ	Absolute Viscosity $[Pa.s]$
ϕ	Volume Fill Fraction [%]
ρ	Density $[kg/m^3]$
ω	Angular Velocity $[{}^{rad}/_{s}]$
Subscripts	
c	Critical
Abbreviations	
atm	Atmosphere
CFD	Computational Fluid Dynamics
HTPB	Hydroxyl Terminated PolyButadiene
RPM	Revolutions Per Minute
SAE	Society of Automotive Engineers
VOF	Volume of Fluid

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II. Introduction

THE increasing demand for space utilization - for applications in satellite communications, earth observation, positioning, research, tourism, and manufacturing - brings a need for cheaper, non-toxic, and more sustainable in-space propellants for satellites to maneuver in orbit and to enable deorbiting at the end of life.

Hybrid propulsion is quite desirable as it combines advantages of both solid propulsion and liquid propulsion systems. Like in the case of a solid propulsion system, hybrid propulsion exhibits architectural simplicity, and like in the case of a liquid propulsion system, it is safer as it can be throttled, stopped and restarted [1–3]. There have been various hybrid propellants that have been researched and developed in the past. They include hydrogen peroxide used with polyethylene, rubber-based fuel used with liquid oxygen, coal used with nitrous oxide, and Hydroxyl Terminated PolyButadiene, HTPB used with nitrous oxide [4]. But perhaps the most successful hybrid propellant is HTPB, mostly used with nitrous oxide, which has been used to successfully power SpaceShipOne; the spacecraft that won the Ansari X-prize in 2004. Further research and development of SpaceShipOne led to the birth of SpaceShipTwo; Virgin Galactic's* suborbital spacecraft designed for space tourism [4].

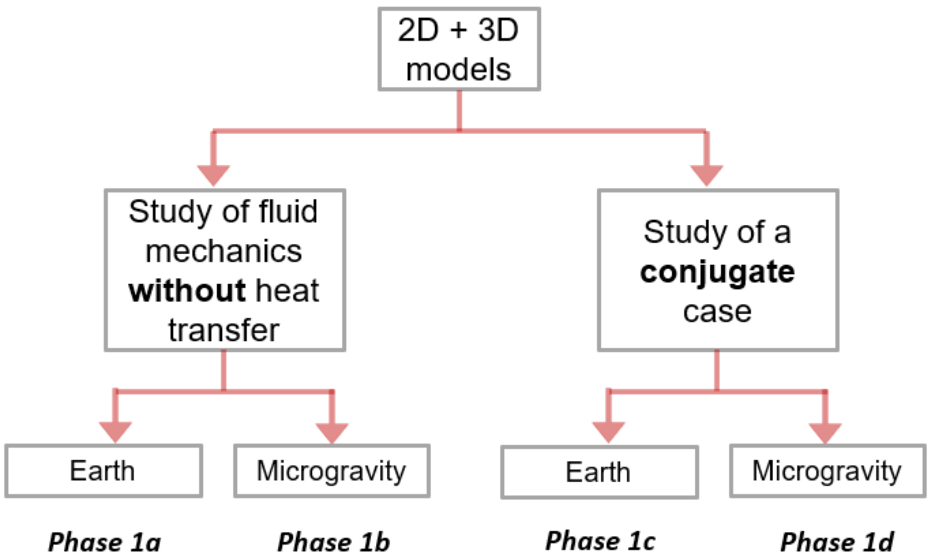

Fig. 1 Work plan summary of the ongoing numerical simulation of centrifugal casting of paraffin wax

Attractive characteristics of paraffin wax (non-toxic, non-hazardous, reusable, cheap, and easy to store) has led to the on-going research and development of paraffin wax as an alternative hybrid propellant [3, 5–8]. Another attractive characteristic of paraffin wax as a hybrid propellant is its high regression rate; which has been found to be around 3 to 5 times greater than the conventional hybrid propellants like HTPB [9]. In addition, the fact that paraffin wax has already being used as a thermal insulator onboard numerous satellites, makes it a very attractive hybrid propellant for in-space propulsion as it could possibly be repurposed for deorbiting purposes at the end of a satellite mission.

The Space Enabled Research Group at the Massachusetts Institute of Technology is pursuing multiyear research on paraffin wax (and other wax-based propellants such as beeswax) as a potential hybrid propellant for launch and in-space propulsion for small-satellite missions. Paraffin wax fuel gains are commonly produced using a centrifugal casting process. Centrifugal casting is a crucial manufacturing process; contributing 15%, in tonnage, of the total castings in the world [10]. Aside from the application of the centrifugal casting process for manufacturing rocket fuel grains, it is also employed in the fabrication of a wide range of products including nanowires [11], reactor fuels [12], machine components such as rotors and gear blanks [13], and composite tubes [14].

The present research focuses on centrifugal casting of rocket fuel grains. Centrifugal casting has long been used for the creation of wax-based hybrid fuel grains. Piscitelli et al. [6, 7] and Stober et al. [15–18] have most recently demonstrated through experiments that centrifugal casting is an effective method of producing hollow casts. Some of the advantages of centrifugal casting over other methods include: better rigidity and surface finish compared to drip casting, and the feasibility of employing centrifugal casting in space.

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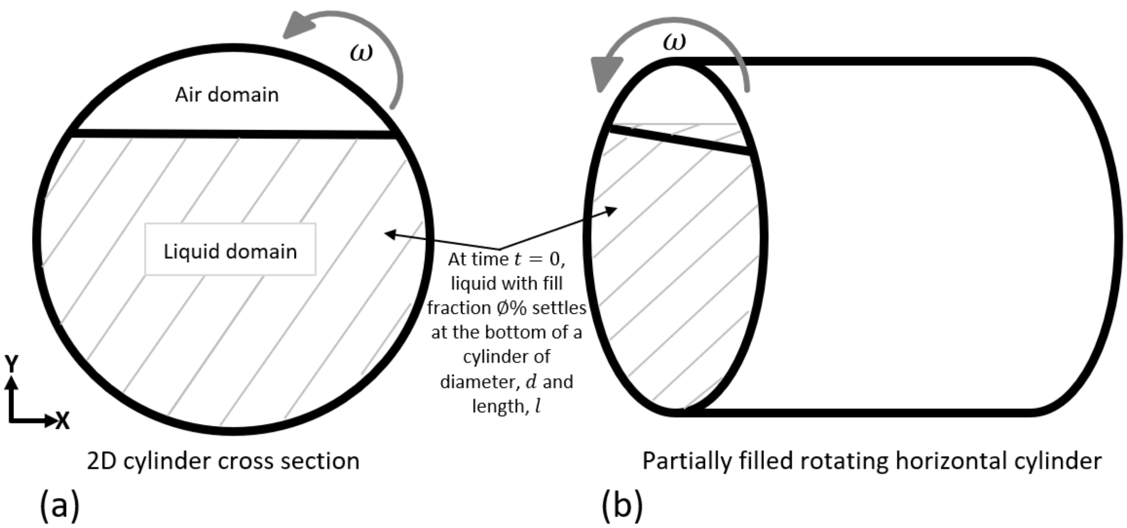

Fig. 2 2D and 3D simplified computational setup

In the centrifugal casting process, liquid is poured into a rotating centrifugal casting tube to partially fill it. As the tube rotates, centrifugal force spins high-density fluid towards the wall of the tube whereas low-density fluid (air) migrates towards the centre of the tube forming a thin-walled hollow cast after solidification. In order to further understand and predict various heat and mass flow phenomena, and even optimise the centrifugal casting process, it is necessary to study, both through experiments and numerical simulations, fluid flow patterns driving the casting process.

Various studies have been conducted in the past to understand the centrifugal casting process. Stober et al. [15–18] have carried out extensive research to study centrifugal casting of paraffin wax both on Earth and in microgravity. In their experiments, they considered three different liquids - water, liquid paraffin wax, and the Society of Automotive Engineers (SAE) 5W-30 motor oil - in order to explore and understand some of the fundamental fluid mechanics and heat transfer mechanisms leveraged in the centrifugal casting process. In their experiments, they also attempted to find the minimum rotational speed needed to form a sound annulus for a given centrifugal casting setup. By running their experiments at a range of rotational speeds, measured by revolutions per minute, RPM they produced time-RPM curves that could be used to estimate the critical rotation speed required to form an annulus for a given centrifugal casting setup.

Rotational speed is one of the crucial parameters that influence the quality of a cast product from the centrifugal casting process. A rotational speed that is lower than the critical casting speed causes liquid material to slip and hence does not adhere to the casting tube walls, causing a phenomenon that some researchers have termed 'raining'[19]. On the other hand, a rotational speed that is higher than the required casting speed may cause hot cracking in the resulting cast product [20]. Jain suggests that in the case of metal molds, the speed of the rotating mold should be sufficient to produce a radial force that is 60-80 times the force of gravity [20]. From a survey of past research, it is evident that critical rotational speed, N_c necessary for a complete annulus formation, in a given centrifugal casting setup, is dependent on various factors such as geometrical parameters of the casting tube, and properties of liquid being rotated.

Experimentally investigating centrifugal casting of metal alloys inside a 0.078 m diameter mold at four different rotational speeds (N = 1097 RPM, N = 1300 RPM, N = 1386 RPM, and N = 1444 RPM), El-Sayed found that a critical rotation speed of N_c = 1097 RPM was needed to produce a uniform annulus [21]. While recently, in their numerical simulation of molten steel rotating in a mold of diameter d = 0.2 m, Barron et al. [19] noted that in their case the critical rotation speed was N_c = 735 RPM.

The aim of the present work was to numerically investigate, using finite volume code of the CFD simulation software ANSYS[†] Fluent, the influence of rotational speed, and fluid viscosity in the centrifugal casting process of paraffin wax. The operating conditions are like those on Earth and with 1 g. A transient two-phase isothermal case, i.e ignoring solidification, was considered. Section III describes the methodology employed for the present study including the details of the computational setup, computational mesh, computational solver and models, and boundary conditions used. Section IV contains the results and discussion while Section V gives the conclusion.

The present work forms the initial phase of an extensive series of numerical work that is currently underway at the

[†] www.ansys.com

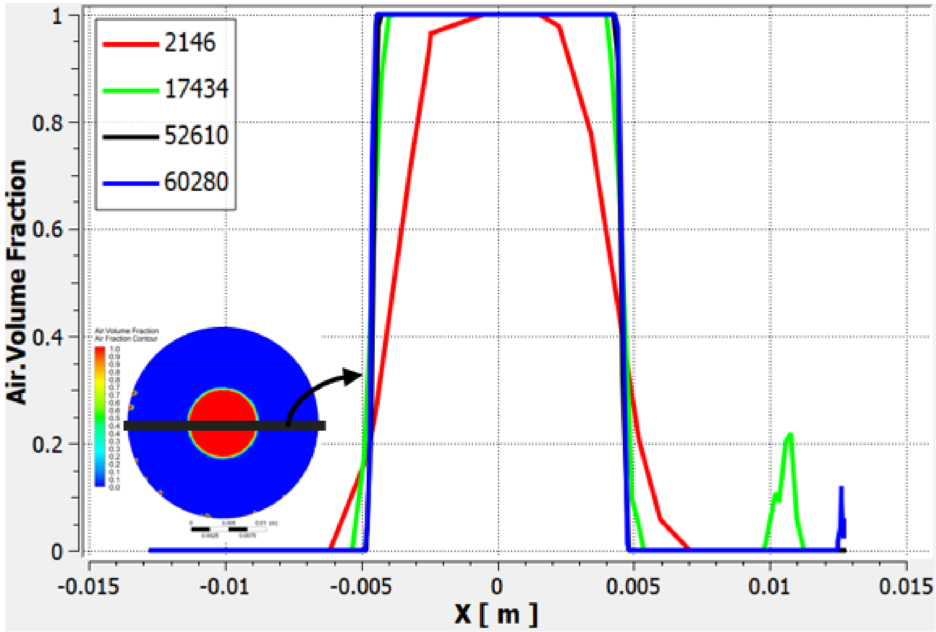

Fig. 3 Mesh sensitivity analysis - radial air volume fraction profile

Space Enabled Research Group at MIT. Fig. 1 shows a summary of the numerical simulation work plan. The numerical work, aside from validating the experiments, will facilitate further understanding of the fluid mechanics and heat transfer mechanisms of the centrifugal casting process. The overarching aim being to better understand centrifugal casting so as to be able to produce better fuel grains both on Earth and in space. Furthermore, as noted before, centrifugal casting is employed in a wide range of industrial applications. Therefore, the knowledge from the present project might help to improve quality of other products that are fabricated via this process.

III. Methodology

A. Computational Setup

Axial symmetry of the partially filled rotating three-dimensional, 3D cylinder was assumed. As such, a two-dimensional, 2D crossectional slice was considered to be a reasonable approximation of the 3D rotation case. The 2D geometry that was used in the present numerical simulation comprised a 2D planar circular crossection of a horizontal partially filled tube. Fig. 2 (a) and (b) show simplified schematic diagrams of the 2D and 3D models respectively. As explained by Stober et al. [15], paraffin wax ($C_{32}H_{66}$) undergoes a shrinkage of about 13.3% during solidification. Despite the fact that solidification was not considered in the present study, to simulate a more practical scenario, the shrinkage factor, of 13.3%, was applied to all the simulation cases studied here. Fig. 2 (a) shows the 2D computation setup with the liquid settled at the bottom and air settled at the top of the cylinder at time t = 0 s, i.e before the start of the simulation.

B. Computational Mesh

An unstructured tetrahedral ANSYS mesh was used. Refinement of the mesh near the walls was done. To ensure that the results were mesh-independent, a mesh sensitivity study was undertaken. The mesh sensitivity study also enabled the selection of an optimum mesh size that offered reasonably accurate simulation results whilst saving on computational time. Four different mesh sizes were considered; 2146, 17434, 52610 and 60280. Radial air volume fraction profiles (Fig. 3) were compared for all the four mesh sizes. A large discrepancy in the simulation results was seen at the lowest mesh count of 2146. The air volume fraction profiles, however, collapsed into almost the same curve as the mesh was further refined. There was not much difference between the results of mesh size 52610 and mesh size 60280. To save on computational time, mesh size of 52610 elements was selected. The mesh quality was judged based

Table 1 Details of the computational solver and models used

Input	Value
Analysis Type	Transient Simulation
Gravity	On
Viscous Model	Standard k-epsilon model
Multiphase Model	Volume of Fluid (VOF)
Pressure-Velocity Coupling Scheme	PISO
Spatial Discretization:	
1. Momentum	Second Order Upwind
2. Volume Fraction	Geo-Reconstruct
3. Pressure	PRESTO!
4. Turbulent Dissipation Rate	Second Order Upwind
5. Turbulent Kinetic Energy	Second Order Upwind

on the mesh quality parameters - for instance skewness - recommended in the ANSYS manual [22].

C. Computational Solver and Models

The present numerical simulations were undertaken using the student version of the CFD software ANSYS Fluent 2020 R1. All the simulations were transient, permitting results visualisation with time. Turbulence was modelled using the Standard k-epsilon viscous model while pressure-velocity coupling was achieved using PISO discretization, which is recommended for transient flow calculations. PRESTO! scheme was chosen for pressure spatial discretization as it is well-suited for steep pressure gradients involved in most rotating flows. PRESTO! provides an improved pressure interpolation where there is a presence of strong pressure variations, for instance in swirling flows. Geo-Reconstruct was chosen for volume fraction spatial discretization as it is the most accurate interface-tracking scheme [22].

There are two main approaches for calculating multiphase flows: the Euler-Lagrange approach, and the Euler-Euler approach. Under these two approaches, are other subcategories of multiphase models. For instance, under Euler-Euler approach there are three main subcategories, namely: mixture model, Volume of Fraction (VOF) model, and Eulerian model. Each of the multiphase models is designed, and is therefore suitable, for a particular multiphase scenario. For a multiphase calculation involving sedimentation, for example, the Mixture model would be the best choice, whereas for cases such as tracking a large bubble inside a liquid, or tracking a liquid-gas interface, VOF model would be the best option [22]. Since the main interest in the present study is to transiently track the position of the interface between two immiscible fluids (liquid and air), Volume of Fluid, VOF model was employed. A summary of the computational solver and models details is given in Table 1.

The VOF multiphase modelling approach has been extensively used in the past by other researchers to sufficiently track a gas-liquid interface. [23–26] employed VOF approach to understand, and predict hydrodynamics of bubbles rising in a vertical column. While [27, 28] used it to transiently analyze thermocapillary behaviour of gas bubbles immersed in a liquid in a microgravity environment. For two or more phases to be modelled using the VOF approach, the phases should not interpenetrate, that is, they should not diffuse into each other [22]. As such, in any cell, based upon the volume fraction values, variables entirely represent one of the phases, or represent a mixture of the phases. For instance, in the present two-phase case, we have air and liquid. Assuming that in a given cell, air volume fraction and liquid volume fraction are denoted as ϕ_{air}, and ϕ_{liquid} respectively. Then, in a VOF approach, there exist three possible conditions, as listed below:

1) if $\phi_{air} = 0$; there is no air in the cell.

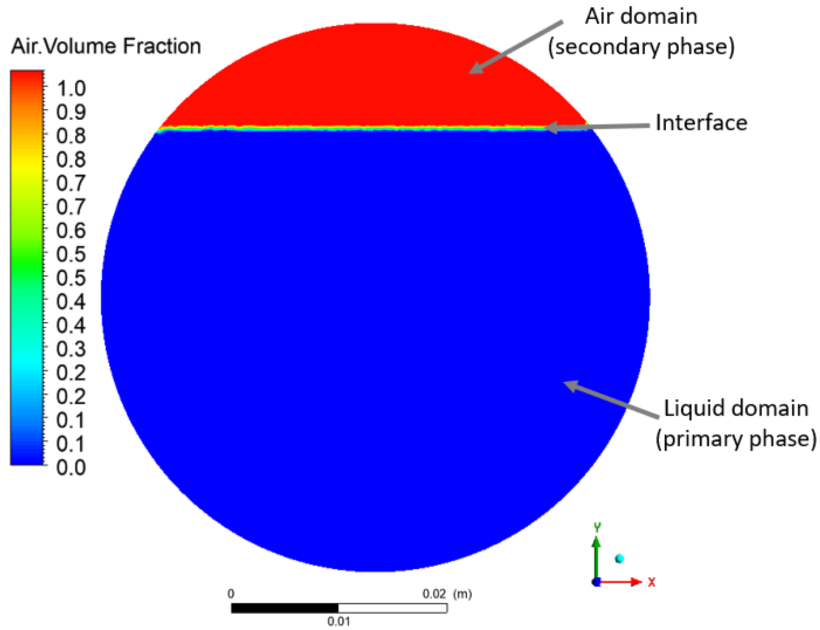

Fig. 4 Initial condition of the primary phase (liquid) and the secondary phase (air) before the start of the transient calculation in ANSYS Fluent

2) if $\phi_{air} = 1$; the cell is full of air.
3) if $0 < \phi_{air} < 1$; the cell has a mixture of air and liquid. This case is common at the air-liquid interface.

Convergence of the simulations was judged on the simulation results no longer changing significantly with an increased number of iterations. Before the calculation began, solution initialization, to provide the solver with an initial 'guess', was done by setting liquid volume fraction to be $\phi_{liquid} = 1$ in the liquid domain and air volume fraction to be $\phi_{air} = 1$ in the air domain, as shown in Fig. 4.

In the present work, as noted before, two parameters; tube rotational speed and fluid viscosity were studied to understand their influence on the centrifugal casting process. Table 2 gives a summary of these parameters as well as their values.

Table 2 A summary of the parameters studied

Parameter	Value
Liquid	Water, Paraffin Wax ($C_{32}H_{66}$), SAE 5W-30 Motor Oil
Rotational Speeds, N	1000 RPM, 2220 RPM, 3000 RPM
Fill Fraction, ϕ	86.5%
Tube Diameter, d	0.0254 m

1. Tube Rotational Speed

In order to estimate an appropriate rotational speed that is not too high as to cause hot cracks on the resulting cast and not too low as to cause 'raining', Jain [20] and El-Sayed [21] proposed that the speed of the rotating mold (their studies focused on metal alloys) should produce a radial force of the molten liquid which is equal to 60 to 80 times the force of gravity. This ratio of centrifugal force to gravitational force is called G-Factor, GF [19, 21]. Centrifugal force is given by Eq. 1, and GF can be expressed as shown in Eq. 2. Substituting velocity ($v = \pi r N/30$) into Eq. 2 and making rotational speed, N, the subject of the equation, results in an equation - Eq. 3 - used for estimating rotational speed needed to produce a sound cast in centrifugal casting of metal casts [19, 21].

$$F = \frac{mv^2}{r} \tag{1}$$

$$G-Factor, GF = \frac{v^2}{rg} \tag{2}$$

$$N = \frac{30}{\pi}\sqrt{\frac{2gGF}{d}} \tag{3}$$

Considering a G-Factor of 60 to 80 proposed by [20], then from Eq. 3, the critical rotational speed needed to sufficiently spin molten metal into a hollow cylinder in a 0.0254 m diameter tube (as used in the present study) is about N_c = 2055 RPM to N_c = 2374 RPM. Assuming an average G-Factor of 70, an approximate critical rotational speed to form an annulus becomes N_c = 2220 RPM. Note that the present simulation used a cylinder diameter of d =0.0254 m that was partially filled to ϕ = 86.5% - as shown in Table 2.

To investigate the influence of rotational speed on the centrifugal casting process, two additional rotational speeds, one lower than the approximate critical rotation speed; N = 1000 RPM, and one higher; N = 3000 RPM. Each of the three liquids - water, liquid paraffin wax, and SAE 5W-30 motor oil - was studied at the three different rotational speeds; N = 1000 RPM, N = 2220 RPM, and N = 3000 RPM. During these simulations, all other parameters, such as tube diameter and volume fill fraction, were kept constant at d =0.0254 m and ϕ = 86.5% respectively.

Table 3 Water, and SAE 5W-30 motor oil viscosity and density values at 20°C. And paraffin wax viscosity and density values at 100°C

Liquid parameters			
Liquid	Density [kg/m^3]	Dynamic Viscosity [mPa.s]	Kinematic Viscosity [m^2/s]
Water	998	1.003	1.01×10^{-6}
Paraffin, Dotriacontane $C_{32}H_{66}$	770	3.73	4.84×10^{-6}
SAE 5W-30 Motor Oil	860	114.7	1.33×10^{-4}

2. Fluid Viscosity

To investigate the influence of viscosity on centrifugal casting, three different liquids (water, SAE 5W-30 motor oil, and liquid paraffin wax $C_{32}H_{66}$) were considered. The fluid properties including viscosity and density of these three liquids are as outlined in Table 3. Stober et al. [15] provides the values for the three liquids at 20°C and 100°C. In the present work, fluid property values at 20°C were used. It was important to consider the temperature of 20°C so as to permit comparison with the experimental results conducted in the lab at room temperature. It should be noted, however, that since paraffin wax is solid at room temperature of 20°C, its viscosity and density values were taken at 100°C.

IV. Results and Discussion

A 2D crossectional slice of a partially filled rotating horizontal cylinder was numerically studied in the finite volume code in the student version of the CFD solver ANSYS Fluent 2020 R1. The influence of rotational speed and fluid viscosity on the centrifugal casting process were investigated following the methodology detailed in Section III of this paper. The simulations focused on a transient two-phase isothermal case. Since the simulations were transient, it was possible to visualize the evolution of annulus formation using contour plots of volume fraction.

The following subsection IV.A and subsection IV.B give more observation and discussion on the influence of rotational speed and fluid viscosity on the centrifugal casting process.

A. Tube Rotation Speed

In order to compare the numerical simulation results with the experimental results of Stober et al. [15], three rotational speeds corresponding to 1000 RPM, 2220 RPM, and 3000 RPM were studied considering a case of SAE 5W-30 motor oil, ϕ = 86.5% and d = 0.0508 m. Stober et al. [15] produced graphs that gave an estimated critical rotation speed for a complete annulus formation in a given centrifugal casting setup. They noted that the rotational

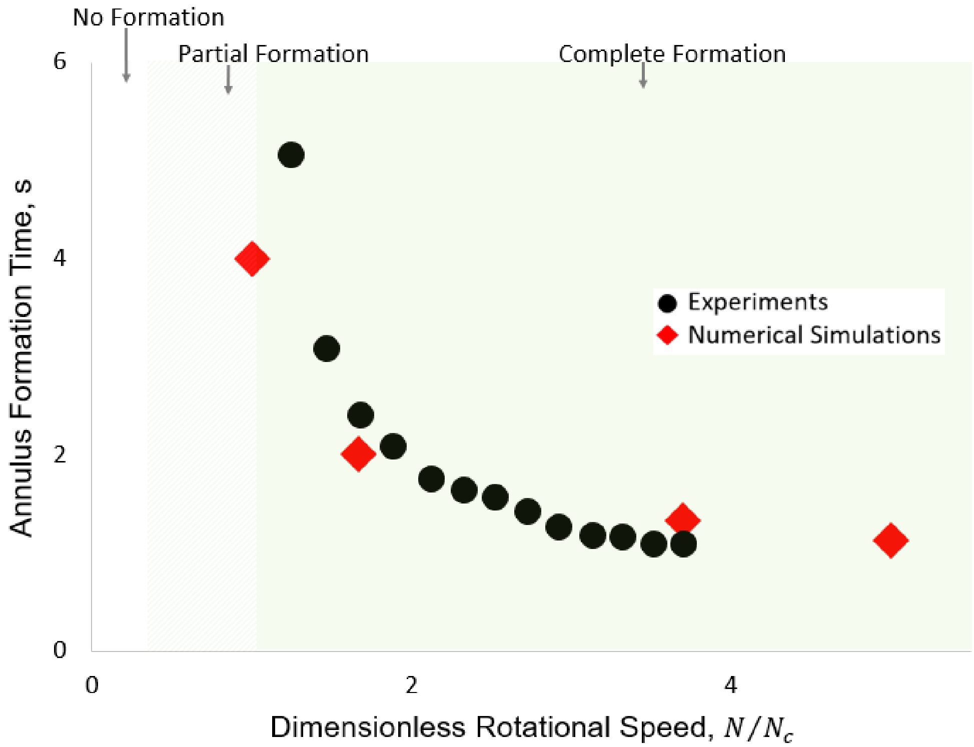

Fig. 5 SAE 5W-30 motor oil annulus formation curve from numerical simulation compared to experimental results of [15]

speeds fell in three regions; no annulus formation region, partial annulus formation region, and complete annulus formation region, as shown in Fig. 5. The rotational speed on the x-axis has been normalized by dividing a given RPM by the critical RPM, N_c. Where the critical RPM, N_c is the minimum RPM necessary for annulus formation. For a case of SAE 5W-30 motor oil rotating in a tube d = 0.0508 m, L = 0.254 m, ϕ = 86.5%, [15] found critical RPM to be N_c = 600. Numerical results for a similar setup, i.e. d = 0.0508 m, ϕ = 86.5% were plotted on the same graph of [15], as shown in Fig. 5.

A similar annulus formation curve can be seen from both the experimental results and the present numerical results. However, the numerical simulation seems to have slightly underestimated the annulus formation time at low rotational speeds. For instance, at $^{N}/_{N_c}$ = 1.67 corresponding to N = 1000 RPM, numerical simulation underestimated the time needed to form an annulus by about 18%. Stober et al. [15] noted that rotational rate threshold for a reliable smooth annulus formation for a case of d = 0.0508 m, L = 0.254 m, ϕ = 86.5% was N_c = 600 RPM. While for the case of d = 0.0508 m, L = 0.102 m, ϕ = 86.5% was N_c = 480 RPM. It seems, therefore, that the time taken to form an annulus is also dependent on the length of the casting tube. This may explain the slight discrepancies that were observed between the present simulation results and the experimental results. [15] used a 3D cylindrical tube of length, L = 0.254 m while the present numerical results considered a 2D geometrical model.

As noted before, to estimate appropriate rotational speed that is not too high as to cause hot cracks on the resulting cast and not too low as to cause 'raining', Jain [20] proposes that the speed of the rotating tube should produce a radial force of the molten liquid which is equal to 60-80 times the force of gravity in case of metal molds. In the present numerical simulations, the proposal by [20] was employed to initially estimate appropriate rotational speed that would produce a sound annulus. However, employing the formula of [20] to estimate critical RPM for a case of SAE 5W-30 motor oil in a cylinder with diameter, d = 0.0508 m and assuming average G-Factor, GF of 70, we get N = 1570; a number approximately 10 times higher than that observed through experiments by [15]. Insinuating that in order to use the equation of estimating critical RPM proposed by [20], GF should be a number lower than that proposed for metal alloys. Dividing the GF of [20] by a factor; which in the present case was found to approximate the ratio of density. For instance, if an average GF for a given metal alloy, say steel is 80 [20], then an approximate GF for estimating the critical rotational speed of SAE 5W-30 motor oil will be $GF \times (^{\rho_{steel}}/_{\rho_{oil}}) \approx 10$. It should be noted, however, that this

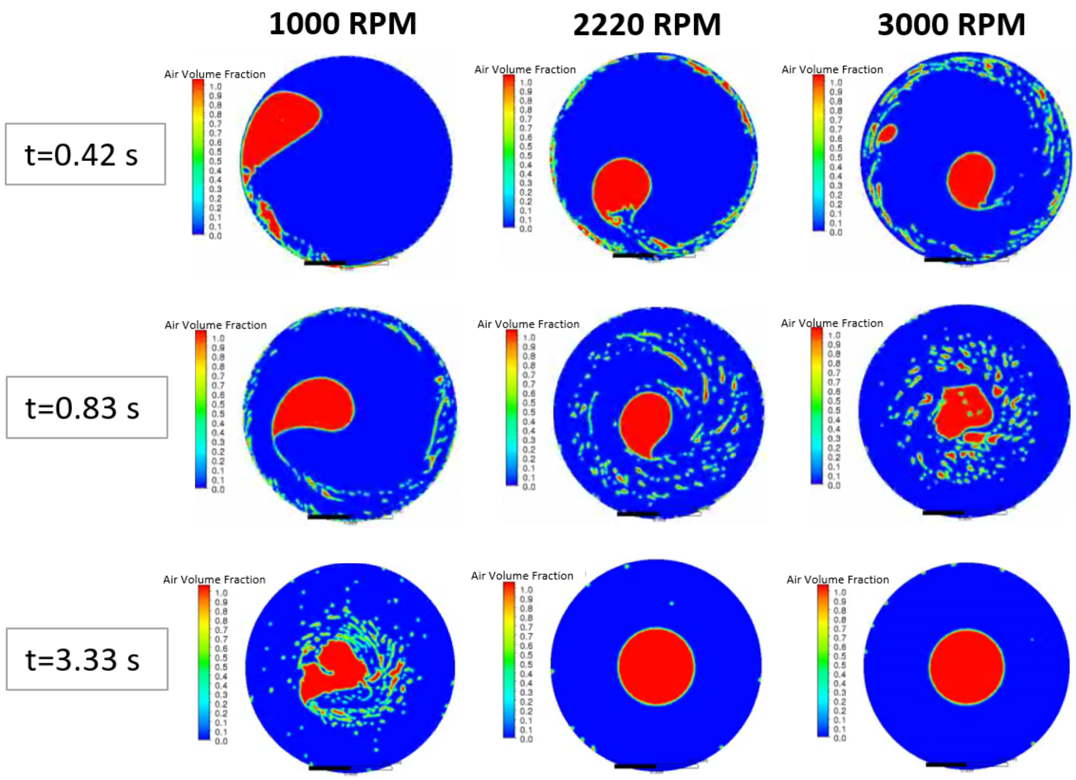

Fig. 6 Air volume fraction contours showing evolution of annulus formation of water for three rotational speeds corresponding to N=1000 RPM, 2220 RPM, and 3000 RPM.

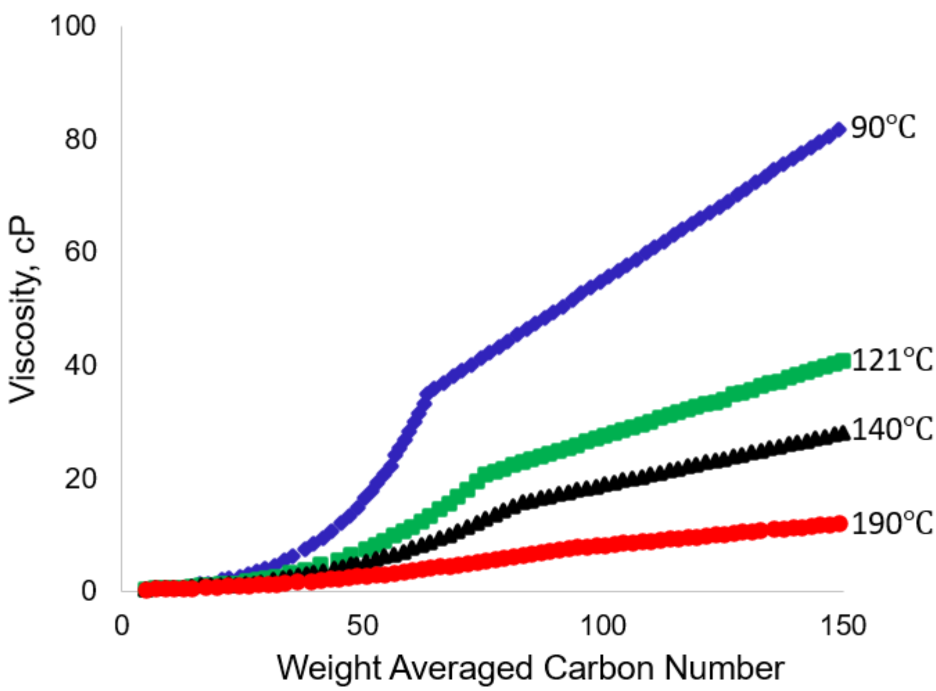

Fig. 7 Alkane viscosity variation with temperature and carbon number, adapted from [29]

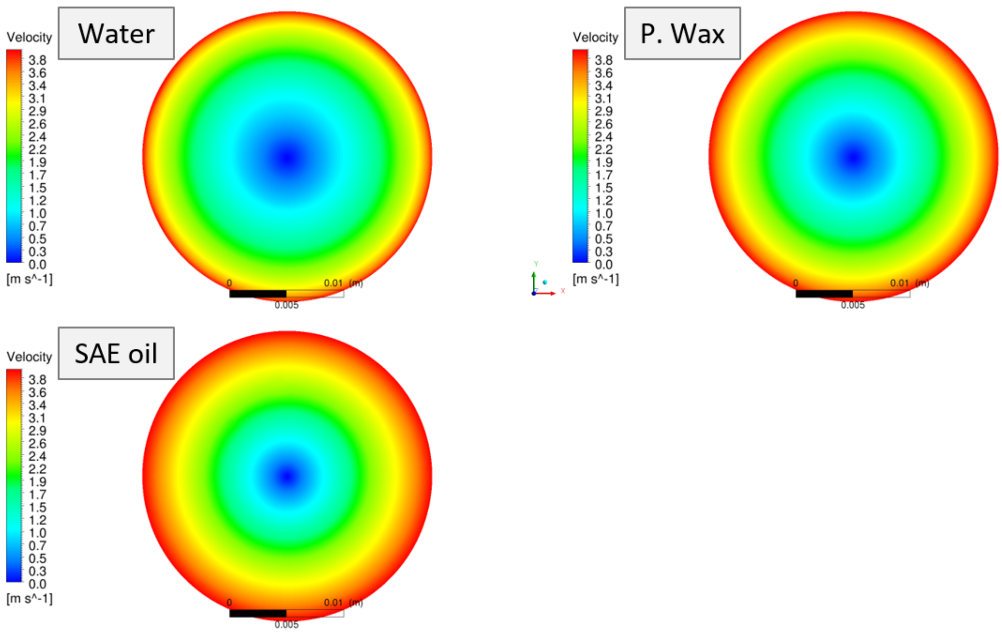

Fig. 8 Water, paraffin wax and SAE 5W-30 motor oil velocity contours at 3000 RPM.

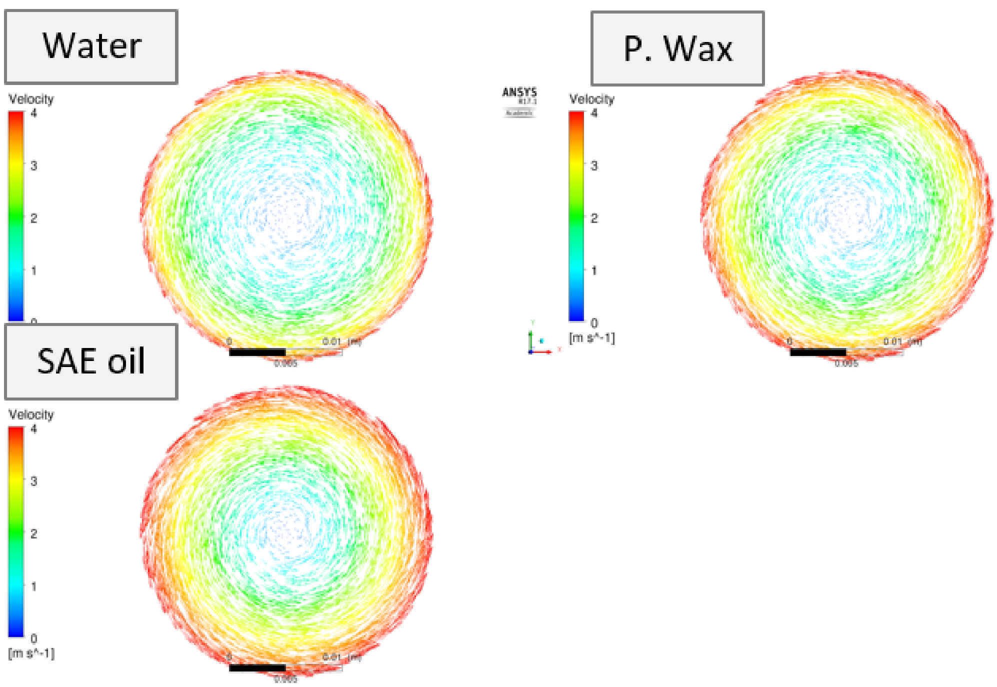

Fig. 9 Water, paraffin wax and SAE 5W-30 motor oil velocity vectors at 3000 RPM.

observation is not conclusive as it only used a single case point. Further investigation into this will be done in future simulation work.

It should be noted that in the present simulations, for any given setup, rotational speed was kept constant throughout the simulation. As such, it was possible to compare the evolution of annulus formation for different rotational speeds at different times of transient simulation. Refer to Fig. 6 for an example of the evolution of annulus formation (visualized by a plot of air volume fraction contours) for a case of water at three rotational speeds corresponding to N = 1000 RPM,

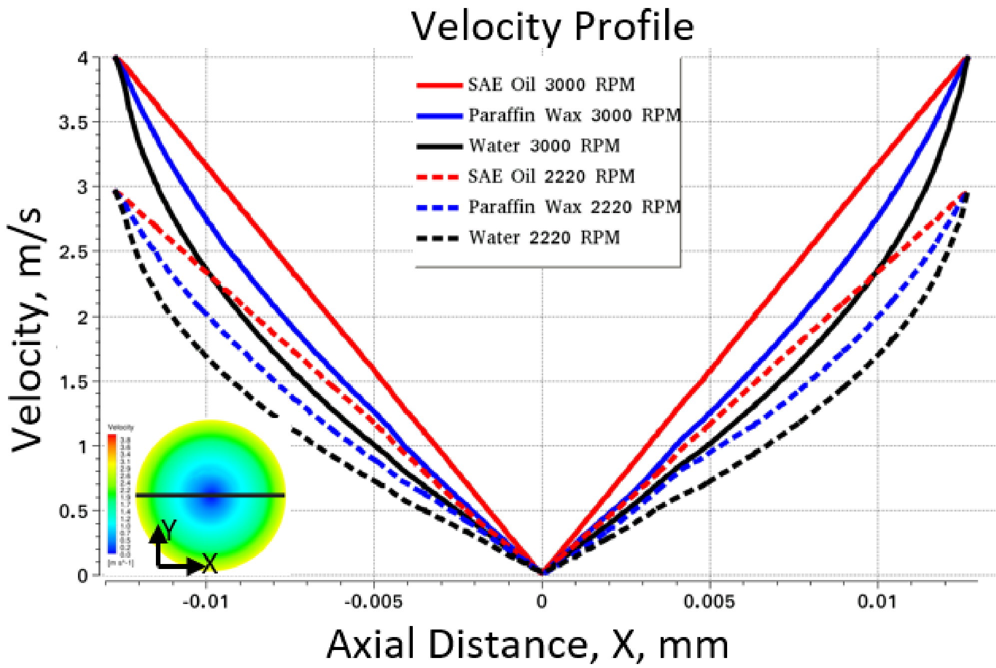

Fig. 10 Water, SAE 5W-30 motor oil and paraffin wax velocity profiles for 2220 RPM and 3000 RPM.

N = 2220 RPM, and N = 3000 RPM. As shown in Fig. 6, the higher the RPM, the faster it was to form a complete annulus. So that a complete annulus was formed by t = 3.33 s if rotational speed was set to be N = 2220 RPM or N = 3000 RPM. Whereas, a relatively low rotational speed of N = 1000 RPM could not produce a complete annulus by t = 3.33 s.

B. Fluid Viscosity

The choice of liquids - water and SAE 5W-30 motor oil - tabulated in Table 3 was informed by the fact that viscosity of paraffin wax, just like any n-alkane, is a factor of two major variables; the number of carbon atoms in the n-alkane, and temperature. Fig. 7 is a graph, adapted from [29], showing how viscosity of n-alkane varies with temperature and number of carbon atoms. In general, viscosity of paraffin wax decreases with an increase in temperature, and increases with an increase in the number of carbon atoms of the n-alkane. Thus, it was necessary to understand fluid mechanics of centrifugal casting at different viscosity values by considering a liquid with high viscosity (e.g SAE 5W-30 motor oil), and a liquid with low viscosity (e.g water). The three liquids - water, SAE 5W-30 motor oil, and liquid paraffin wax - exhibited varied fluid flow behaviour as depicted by plots of velocity contours, velocity vectors, velocity profile, and air volume fraction contours in Fig. 8, Fig. 9, Fig. 10, and Fig. 11 respectively.

Viscosity is a measure of the internal forces of friction in a fluid. Therefore, it has a direct influence on viscous drag; a key parameter that influences the flow phenomenon in a centrifugal casting process [30]. Other key parameters include fluid inertia, surface tension, and gravity. While surface tension might have insignificant effect on centrifugal casting on Earth [30], this might not be the case in a low-gravity environment. It is expected that in a microgravity environment, surface tension will be one of the additional key factors influencing fluid mechanics of centrifugal casting. Researchers through computer simulations [27, 28], and through experiments at the International Space Station [31], investigating thermocapillary behavior of gas bubbles in a microgravity environment, found that surface tension played a key role in gas-liquid interaction in a microgravity environment. Causing fluids to exhibit anomalous behavior that is significantly different from that observed on Earth [31].

High viscosity liquids like SAE 5W-30 motor oil attained an annulus earlier and at relatively lower rotational speeds, compared to low viscosity liquids like water. See Fig.11 which shows the evolution of annulus formation, visualized using contours of air volume fraction, for the three liquids; water, liquid paraffin wax, and SAE 5W-30 motor oil at N = 3000 RPM. Even though a complete annulus was achieved in the end for all the three liquids, SAE 5W-30 motor oil formed an annulus faster than paraffin wax and water - by around t =1.25 s.

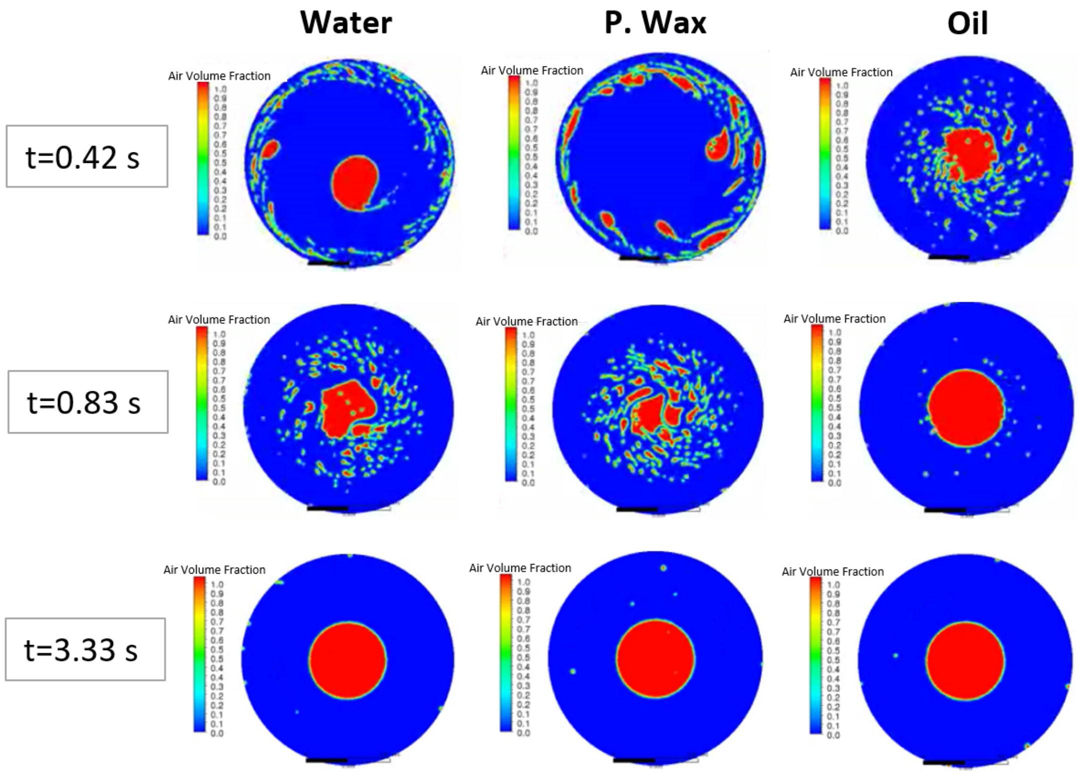

Fig. 11 Air volume fraction contours showing evolution of the distribution of Water, paraffin wax and SAE 5W-30 motor oil for a rotational speed corresponding to N=3000 RPM.

Owing to high viscosity in liquids like SAE 5W-30 motor oil, more layers of the liquid experience velocity of the tube walls unlike less viscous liquids like water which have a very thin liquid layer experiencing tube wall velocity - see Fig. 8. Therefore, unlike paraffin wax and water velocity profiles, the SAE 5W-30 motor oil velocity profile closely approximated that of a solid body rotation - see Fig. 10. Implying that the more viscous the fluid, the more its velocity profile approximates that of a rigid body. This observation is in line with what Thoroddsen and Mahadevan noted; in the event where centrifugal forces dominated, liquids coated the rotating tube uniformly and rigidly rotated with it [32].

The dynamic viscosity of SAE 5W-30 motor oil is about 114 times that of water while its density is only about 0.86 times that of water, see Table 3. Implying that the main determinant of the differences noted in the fluid flow mechanics of the three liquids is viscosity. Other researchers have observed this same influence of viscosity in the centrifugal casting process. Keerthiprasad et al. undertook both numerical simulations and experiments to investigate the influence of liquid viscosity on centrifugal casting [33]. They noted that viscosity played a vital role in the centrifugal casting process; viscous fluid required lower rotational speeds to form an annulus, compared to less viscous flow like water. They further noted that sloshing and other phenomena like Ekman flow and turbulence were observed only for low-viscosity liquids. These phenomena were negligible in high-viscosity liquids like motor oil.

The ability for more viscous liquids to form an annulus easier than less viscous liquids is confirmed by effective angular velocity, which Thoroddsen and Mahadevan defined as $\phi^2 g/r\mu$ [32]. Thoroddsen and Mahadevan noted that the coating capability of liquids scaled with the effective angular velocity [32]. This means that for the same volume fill fraction, ϕ tube radius, r same g, a big value of μ (for instance, in the case of SAE 5W-30 motor oil) causes coating to be achieved at relatively lower effective angular velocity.

Once the simulations on the 2D slice of the casting tube is complete, numerical study will be extended to a 3D geometry. See Fig. 2 (b) for a simplified diagram of the 3D geometry of a partially filled centrifugal casting tube that will be used. The same procedure of numerical analysis, as outlined in the methodology section of this paper, will be followed. In the case of a 3D geometry study, owing to a relatively larger mesh size, computational time is expected to significantly increase. In addition, it is anticipated that the results will closely match that of a 2D geometry reported in this paper. However, some discrepancies are expected to arise mainly as a result of the influence of Ekman flow phenomenon; end flow that is inevitable in a tube with a finite length. As Keerthiprasad et al. [33] has hinted, the Ekman flow is anticipated to be more pronounced in low-viscosity liquids like water and less pronounced in high-viscosity liquids like SAE 5W-30 motor oil.

Much later, numerical simulations will be undertaken for a conjugate scenario where fluid flow mechanics and heat transfer and cooling are calculated simultaneously. In this case of rotation with heat transfer, since heat is being conducted away during the simulations, it is expected that time needed to form an annulus will be different from that of a case without solidification. Furthermore, due to varying fluid properties with time, as solidification process takes place, it is anticipated that fluid flow mechanics visualized using plots of velocity contours, velocity profiles, and air volume fraction contours will differ from those obtained in the present case of fluid rotation without solidification.

V. Conclusion

Transient two-phase isothermal numerical simulations in the finite volume code of the student version of the CFD solver ANSYS Fluent 2020 R1 were undertaken to study the influence of rotational speed and fluid viscosity in the centrifugal casting process of paraffin wax. Volume of Fluid (VOF) model proved to be a useful tool in studying transient simulation of two-phase fluid rotation. The VOF model was able to transiently track the position of the interface between two immiscible, non-penetrating fluids. Permitting evolution of annulus formation to be visualized using plots of air volume fraction contours. The influence of fluid viscosity and rotational speed were studied to understand their influence on the centrifugal casting process. Three liquids; water, SAE 5W-30 motor oil, pa1raffin wax, and three rotational speeds; 1000 RPM, 2220 RPM, and 3000 RPM were considered.

High viscosity liquids like SAE 5W-30 motor oil formed an annulus even at relatively low RPM. Whereas low viscosity liquids like water only formed a smooth reliable annulus at relatively higher rotational speeds of 3000 RPM. Furthermore, the higher the rotational speed used, the faster it was to form a complete annulus. The numerical simulation results compared fairly well, more so at relatively higher rotational speeds. At relatively lower RPM, however, numerical simulations were found to slightly underestimate annulus formation time - for instance, at rotation speeds of of 1000 RPM, simulations underestimated the time needed to form an annulus by about 18%.

Future work will involve repeating the same numerical simulation study but considering a more practical case using 3D geometry. It should be noted that, the research presented here focused on fluid mechanics only, that is, heat transfer and cooling were not factored in the simulations. Future work will also extend numerical analysis so as to study fluid mechanics and heat transfer simultaneously. Since the ultimate aim is to use wax-based propellants for launch and in-space propulsion of small satellites - e.g., deorbiting satellites at the end of life - part of the future work will also include numerical analysis to obtain insights into the heat and mass transfer phenomena of wax-based propellants in microgravity conditions.

Acknowledgments

The authors would like to sincerely thank: (1) the Schmidt Science Fellows for financially supporting the lead author. (2) Massachusetts Institute of Technology, MIT Gas Turbine Labaratory (GTL) for granting the present authors permission, and guidance, to run the numerical simulations in their computational clusters.

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